Applied Fluid Mechanics

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Applied Fluid Mechanics (A07182)

Short Title:	Applied Fluid Mechanics		
Faculty	Engineering		
Department	Engineering Technology		
Cluster	Engineering		
Author	GGALLAGHER		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module builds on the fundamentals of fluids established in the Fluid Mechanics module in year 3 of this programme. The approach is to look at real fluids in real applications. It further develops the principles underlying the subject and demonstrates how these are used for the design of hydraulic components and systems.

Programmes

FPOW-0005	BEng (Hons) in Mechanical and Manufacturing Engineering (WD_EMSEN_B)	4 / 8 / M
FPOW-0005	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 8 / E

Indicative Content

- Review of basic fluid properties. Fluid statics. Bernoulli and continuity equations.
- Behaviour of real fluids. Flow in bounded systems
- Flow in pipelines. Friction effects. General energy equation. Pump sizing and selection
- External flow: boundary layer, drag and lift
- Dimensional analysis and similarity
- Analysis of fluid flow in rotodynamic machinery such as: centrifugal pump, Francis turbine and Pelton wheel

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Design simple fluids systems accounting for the various energy losses the fluid experience and select a suitable pump.
2. Explain the origin of lift and drag in a variety of external flow scenarios and evaluate consequent drag and lift forces.
3. Apply dimensional analysis and similarity techniques.
4. Analyse simple fluid flows and determine important fluid parameters in such flows.
5. Describe a range of turbomachines and analyse fluid flow in same.

Learning and Teaching Methods

- Learning and teaching methods and strategies:
- 1. Lectures
- 2. Assignments
- 3. Laboratories

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
In-Class Assessment	15%	1,4
Project	15%	1,4

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Independent Learning	87	

Supplementary Material

- Cengel, Y. and J.M. Cimbala. _Fluid Mechanics: Fundamentals and Applications_. USA: McGraw-Hill, 2013.
- Douglas, J.F., J.M. Gasiorek, J.A. Swaffield and L.B. Jack. _Fluid Mechanics_. UK: Prentice Hall, 2011.
- Mott, R.L. _Applied Fluid Mechanics_. USA: Prentice Hall, 2006.
- Potter, M., D.C. Wiggert and B.H. Ramadan. _Mechanics of Fluids_. USA: Cengage, 2011.